The documentation and process conversion measures necessary to comply with this revision shall be completed by 17 Nov 94

INCH POUND

MIL-S-19500/542E
17 August 1994
SUPERSEDING
MIL-S-19500/542D
30 June 1992

MILITARY SPECIFICATION

SEMICONDUCTOR DEVICE, FIELD EFFECT TRANSISTORS, N-CHANNEL, SILICON
TYPES 2N6756, 2N6758, 2N6760, 2N6762,
JANTX, JANTXV, JANS, JANHC, and JANKC

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the detail requirements for N-channel, enhancement-mode, MOSFET, power transistors intended for use in high density power switching applications. Three levels of product assurance are provided for each device type as specified in MIL-S-19500. Two levels of product assurance are provided for each unencapsulated device type. See 6.4 for JAN level.

1.2 Physical dimensions. See figure 1 (TO-204AA; formerly TO-3), figures 2, 3, and (see 6.5) for JANHC and JANKC die dimensions.

1.3 Maximum ratings. Unless otherwise specified, T_A = +25°C.

Type	P_T 1/ T_C = +25°C	P_T T_A = +25°C (free air)	V_{DS}	V_{DG}	V_{GS}	I_{D1} 2/ T_C = +25°C	I_{D2} 2/ T_C = +100°C
	W	W	V dc	V dc	V dc	A dc	A dc
2N6756	75	4	100	100	±20	14.0	9.0
2N6758	75	4	200	200	±20	9.0	6.0
2N6760	75	4	400	400	±20	5.5	3.5
2N6762	75	4	500	500	±20	4.5	3.0

Type	I_S	I_{DM} 3/	T_J and T_{STG}	V_{ISO} 70,000 feet altitude	Max $r_{DS(on)}$ 1/ V_{GS} = 10 V dc, $I_D = I_{D2}$		$R_{\theta JC}$ max
					T_J = 25°C	T_J = 150°C	
	A dc	A (pk)	°C		ohms	ohms	°C/W
2N6756	14.0	56	-55 to +150		0.18	0.36	1.67
2N6758	9.0	36	-55 to +150		0.4	0.84	1.67
2N6760	5.5	22	-55 to +150	400	1.0	2.5	1.67
2N6762	4.5	18	-55 to +150	500	1.5	3.75	1.67

1/ Derate linearly 0.6 W/°C for T_C > +25°C. $P_T = \dfrac{T_J \text{ max} - T_C}{R_{\theta JC}}$

2/ $I_D = \sqrt{\dfrac{T_J \text{ max} - T_C}{(R_{\theta JC} \times R_{DS(on)} \text{ at } T_J \text{ max})}}$

3/ I_{DM} = 4 I_{D1}; I_{D1} as calculated in note 2.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Defense Electronics Supply Center, ATTN: DESC-ELD, 1507 Wilmington Pike, Dayton, OH 45444-5765, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A

FSC 5961

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1.4 Primary electrical characteristics. Unless otherwise specified, T_C = +25°C.

Type	Min $V_{(BR)DSS}$ V_{GS} = 0 V I_D = 1 mA dc	$V_{GS(th)1}$ $V_{DS} \geq V_{GS}$ I_D = 0.25 mA		Max I_{DSS1} V_{GS} = 0 V V_{DS} = 80 percent of rated V_{DS}	Max $r_{DS(on)1}$ 1/ V_{GS} = 10 V dc $I_D = I_{D2}$ T_J = +25°C
	V dc	V dc		μA dc	ohms
		Min	Max		
2N6756	100	2.0	4.0	25	0.18
2N6758	200	2.0	4.0	25	0.4
2N6760	400	2.0	4.0	25	1.0
2N6762	500	2.0	4.0	25	1.5

1/ Pulsed, (see 4.5.1).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATION

MILITARY

MIL-S-19500 - Semiconductor Devices, General Specification for.

STANDARD

MILITARY

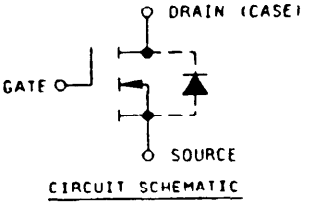
MIL-STD-750 - Test Methods for Semiconductor Devices.

(Unless otherwise indicated copies of federal and military specifications, standards, and handbooks are available from the Defense Printing Service Detachment Office, Building 4D (Customer Service), 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

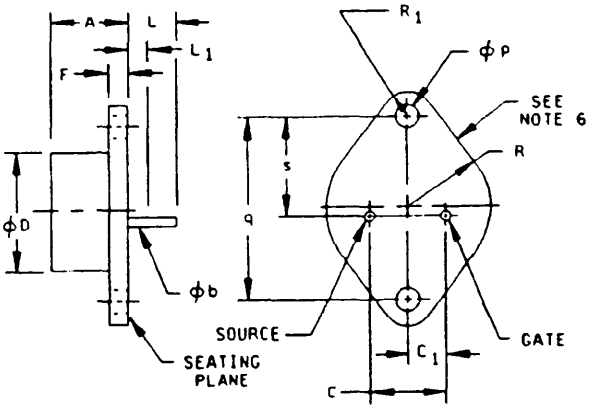
2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

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Symbol	Dimensions				Notes
	Inches		Millimeters		
	Min	Max	Min	Max	
A	.250	.360	6.35	9.14	
ϕb	.038	.043	0.97	1.09	
ϕD		.875		22.23	
C	.420	.440	10.67	11.18	3, 5
C_1	.205	.225	5.21	5.72	3, 5
F	.060	.135	1.52	3.43	
L	.312	.500	7.92	12.70	
L_1		.050		1.27	
ϕp	.151	.161	3.84	4.09	
q	1.177	1.197	29.90	30.40	
R	.495	.525	12.57	13.34	
R_1	.131	.188	3.33	4.78	
s	.655	.675	16.64	17.15	



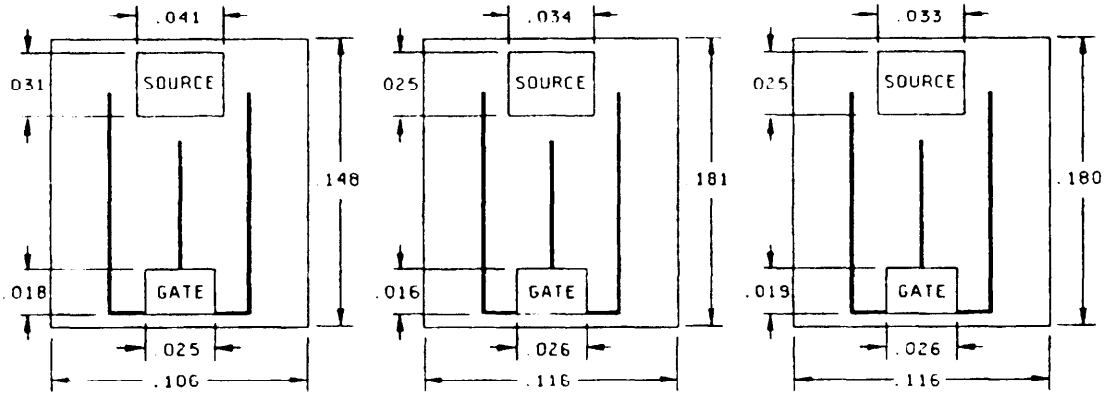
CIRCUIT SCHEMATIC



NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. These dimensions should be measured at points .050 inch (1.27 mm) and .055 inch (1.40 mm) below seating plane. When gauge is not used measurement will be made at the seating plane.
4. The seating plane of the header shall be flat within .001 inch (0.03 mm) concave to .004 inch (0.10 mm) convex inside a .930 inch (23.62 mm) diameter circle on the center of the header and flat within .001 inch (0.03 mm) concave to .006 inch (0.15 mm) convex overall.
5. Mounting holes shall be deburred on the seating plane side.
6. Drain is electrically connected to the case.

FIGURE 1. Physical dimensions of transistor (TO-204AA).

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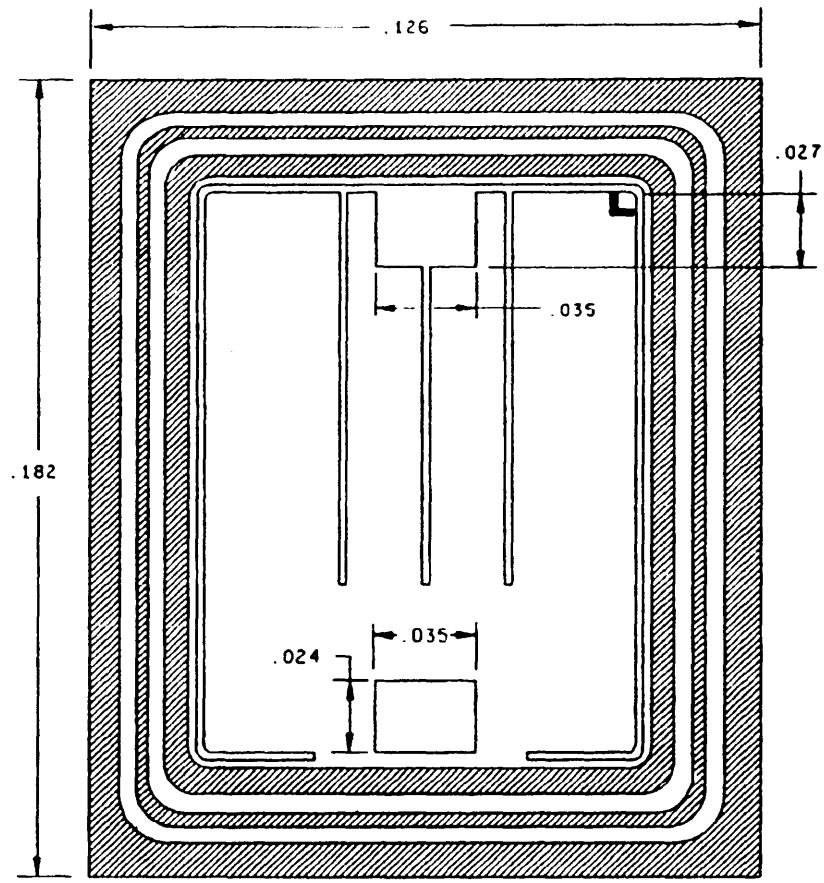


Inch	mm	Inch	mm	Inch	mm
.016	0.41	.031	0.79	.116	2.95
.018	0.46	.033	0.84	.148	3.76
.019	0.48	.034	0.86	.180	4.57
.025	0.64	.041	1.04	.181	4.60
.026	0.66	.106	2.69		

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerance is ±.005 inch (0.13 mm).
4. The physical characteristics of the die thickness are .0187 inch (0.475 mm). The back metals are chromium, nickel and silver. The top metal is aluminum and the back contact is the drain.

FIGURE 2. JANHCA and JANKCA (A-version).

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Inch	mm
.024	0.61
.027	0.69
.035	0.89
.126	3.20
.182	4.62

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerance is ±.005 inch (0.13 mm).
4. The physical characteristics are the die thickness .014 inch (0.36 mm). The back metals are aluminum, nickel, and titanium. The top metal is aluminum and the back contact is the drain.

FIGURE 3. JANHCB and JANKCB (B-version) die dimensions for 2N6756, 2N6758, 2N6760, and 2N6762.

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3. REQUIREMENTS

3.1 Associated detail specification. The individual item requirements shall be in accordance with MIL-S-19500 and as specified herein.

3.2 Abbreviations, symbols, and definitions. Abbreviations, symbols, and definitions used herein shall be as specified in MIL-S-19500.

3.3 Design, construction, and physical dimensions. The design, construction, and physical dimensions shall be as specified in MIL-S-19500, and on figures 1 and 2 herein.

3.3.1 Lead material and finish. Lead material shall be Kovar or Alloy 52; a copper core or plated core is permitted. Lead finish shall be solderable in accordance with MIL-STD-750, MIL-S-19500, and herein. Where a choice of lead material or finish is desired, it shall be specified in the acquisition document (see 6.2).

3.3.2 Internal construction. Multiple chip construction is not permitted to meet the requirements of this specification.

3.4 Marking. Marking shall be in accordance with MIL-S-19500.

3.5 Electrostatic discharge protection. The devices covered by this specification require electrostatic protection.

3.5.1 Handling. MOS devices must be handled with certain precautions to avoid damage due to the accumulation of electrostatic charge. The following handling practices shall be followed:

a. Devices shall be handled on benches with conductive handling devices.

b. Ground test equipment, tools, and personnel handling devices.

c. Do not handle devices by the leads.

d. Store devices in conductive foam or carriers.

e. Avoid use of plastic, rubber, or silk in MOS areas.

f. Maintain relative humidity above 50 percent if practical.

g. Care shall be exercised, during test and troubleshooting, to apply not more than maximum rated voltage to any lead.

h. Gate must be terminated to source, $R \leq 100$ k, whenever bias voltage is to be applied drain to source.

4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection shall be in accordance with MIL-S-19500 and as specified herein.

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-S-19500 and as specified herein. Alternate flow is allowed for qualification inspection in accordance with figure 2 of MIL-S-19500.

4.2.1 Group E. Group E inspection shall be conducted in accordance with MIL-S-19500 and table II herein.

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4.3 Screening (JANS, JANTX, and JANTXV levels only). Screening shall be in accordance with table II of MIL-S-19500 and as specified herein. The following measurements shall be made in accordance with table I herein. Devices that exceed the limits of table I herein shall not be acceptable.

Screen (see table II of MIL-S-19500)	Measurement	
	JANS level	JANTX and JANTXV levels
1/	Gate stress test, (see 4.5.5)	Gate stress test, (see 4.5.5)
1/ 2/	Method 3470, (see 4.5.4) optional	Method 3470, (see 4.5.4) optional
1/	Method 3161, (see 4.5.3)	Method 3161, (see 4.5.3)
9	I_{GSS1}, I_{DSS1}	
10	Method 1042, test condition B	Method 1042, test condition B
11	I_{GSS1}, I_{DSS1}, $r_{DS(on)1}$, $V_{GS(th)1}$ Subgroup 2 of table I herein: ΔI_{GSS1} = ±20 nA dc or ±100 percent of initial value, whichever is greater. ΔI_{DSS1} = ±25 µA dc or ±100 percent of initial value, whichever is greater.	I_{GSS1}, I_{DSS1}, $r_{DS(on)1}$, $V_{GS(th)1}$, subgroup 2 of table I herein.
12	Method 1042, test condition A, t = 240 hrs	Method 1042, test condition A t = 48 hours min at 175°C min, optional
13	Subgroups 2 and 3 of table I herein: ΔI_{GSS1} = ±20 nA dc or ±100 percent of initial value, whichever is greater. ΔI_{DSS1} = ±25 µA dc or ±100 percent of initial value, whichever is greater. $\Delta r_{DS(on)1}$ = ±20 percent of initial value $\Delta V_{GS(th)1}$ = ±20 percent of initial value	Subgroup 2 of table I herein: ΔI_{GSS1} = ±20 nA dc or ±100 percent of initial value, whichever is greater. ΔI_{DSS1} = ±25 µA dc or ±100 percent of initial value, whichever is greater. $\Delta r_{DS(on)1}$ = ±20 percent of initial value $\Delta V_{GS(th)1}$ = ±20 percent of initial value

1/ Shall be performed anytime before screen 10.
2/ Method 3470 is optional if performed as a sample in group A, subgroup 5.

4.3.1 Screening (JANHC and JANKC). Screening of die shall be in accordance with MIL-S-19500 appendix H. As a minimum Die, shall be 100 percent probed per group A, subgroup 2 except test current shall not exceed 20 amperes.

4.3.2 JANHC and JANKC die. Qualification shall be in accordance with MIL-S-19500.

4.4 Quality conformance inspection. Quality conformance inspection shall be in accordance with MIL-S-19500, and as specified herein. Alternate flow is allowed for quality conformance inspection in accordance with figure 3 of MIL-S-19500.

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4.4.1 Group A inspection. Group A inspection shall be conducted in accordance with MIL-S-19500 and table I herein. Electrical measurements (end-points) shall be in accordance with the inspections of table I, group A, subgroup 2 herein.

4.4.2 Group B inspection. Group B inspection shall be conducted in accordance with the conditions specified for subgroup testing in table IVa (JANS) and table IVb (JANTX and JANTXV) of MIL-S-19500 and herein. Electrical measurements (end-points) shall be in accordance with the inspections of table I, group A, subgroup 2 herein.

4.4.2.1 Group B inspection, table IVa (JANS) of MIL-S-19500.

Subgroup	Method	Condition
3	1051	Test condition G.
4	1042	Test condition D; the heating cycle shall be 1 minute minimum, 2,000 cycles. No heat sink nor forced air cooling on the device shall be permitted during the cycle.
5	1042	Accelerated steady-state operation life; test condition A, V_{DS} = rated T_A = +175°C, t = 120 hours minimum. Read and record $V_{(BR)DSS}$ (pre and post at 1 mA = I_D. Read and record I_{DSS} (pre and post). Deltas for $V_{(BR)DSS}$ shall not exceed 10% and I_{DSS} shall not exceed 25μA. Accelerated steady-state gate stress; condition B, V_{GS} = rated, T_A = +175°C, t = 24 hours.
5	2037	Bond strength (Al-Au die interconnects only); test condition A.
6	3161	See 4.5.2.

4.4.2.2 Group B inspection, table IVb (JANTX and JANTXV) of MIL-S-19500.

Subgroup	Method	Condition
2	1051	Test condition G.
3	1042	Test condition D, 2,000 cycles minimum. The heating cycle shall be 1 minute minimum.
3	2037	Test condition A. All internal bond wires for each device shall be pulled separately.
4	2075	See 3.3.2.
6	----	Not applicable.

4.4.3 Group C inspection. Group C inspection shall be conducted in accordance with the conditions specified for subgroup testing in table V of MIL-S-19500 and as follows. Electrical measurements (end-points) shall be in accordance with the inspections of table I, group A, subgroup 2 herein.

Subgroup	Method	Condition
2	1056	Test condition A.
2	2036	Test condition A; weight = 10 lbs, t = 15 s.
6	1042	Test condition D; 6,000 cycles minimum. The heating cycle shall be 1 minute minimum.

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4.5.1 Pulse measurements. Conditions for pulse measurement shall be as specified in section 4 of MIL-STD 750.

4.5.2 Thermal resistance. Thermal resistance measurements shall be performed in accordance with method 3161 of MIL-STD-750. $R_{\Theta JC}$ max = 1.67°C/W.

a.	I_M measuring current	10 mA.
b.	I_H drain heating current	1.5 A.
c.	t_H heating time	Steady-state (see MIL-STD-750, method 3161 for definition).
d.	V_H drain-source heating voltage	20 V.
e.	t_{MD} measurement time delay	30 to 60 µs.
f.	t_{SW} sample window time	10 µs (max).

4.5.3 Thermal impedance ($Z_{\Theta JC}$ measurements). The $Z_{\Theta JC}$ measurements shall be performed in accordance with MIL-STD-750, method 3161. The maximum limit (not to exceed figure 4, thermal impedance curves and the group A, subgroup 2 limits) for $Z_{\Theta JC}$ in screening (table II of MIL-S-19500) shall be derived by each vendor by means of statistical process control. When the process has exhibited control and capability, the capability data shall be used to establish the fixed screening limit. In addition to screening, once a fixed limit has been established, monitor all future sealing lots using a random five piece sample from each lot to be plotted on the applicable $\bar{X}$, R chart. If a lot exhibits an out of control condition, the entire lot shall be removed from the line and held for Engineering evaluation and disposition. This procedure may be used in lieu of an inline process monitor.

a.	I_M measuring current	10 mA.
b.	I_H drain heating current	1.5 A (min).
c.	t_H heating time	100 ms.
d.	V_H drain-source heating voltage	20 V (min).
e.	t_{MD} measurement time delay	30 to 60 µs.
f.	t_{SW} sample window time	10 µs (max).

4.5.4 Single pulse unclamped inductive switching.

a.	Peak current, I_D	rated I_D.
b.	Peak gate voltage, V_{GS}	10 V.
c.	Gate to source resistor, R_{GS}	25 ≤ R_{GS} ≤ 200.
d.	Initial case temperature	+25°C +10, -5°C.
e.	Inductance	100 µH minimum.
f.	Number of pulses to be applied	1 pulse minimum.
g.	Supply voltage V_{DD}	50 V.

4.5.5 Gate stress test.

a. V_{GS} ± 30 V minimum.

b. t = 250 µs minimum.

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TABLE I. Group A inspection.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 1						
Visual and mechanical inspection	2071					
Subgroup 2						
Thermal impedance 2/	3161	See 4.5.3	$Z_{\Theta JC}$		1.6	°C/W
Breakdown voltage, drain to source	3407	V_{GS} = 0 V dc I_D = 1 mA dc Condition C	$V_{(BR)DSS}$			V dc
2N6756				100		
2N6758				200		
2N6760				400		
2N6762				500		
Gate to source voltage (threshold)	3404	$V_{DS} \geq V_{GS}$ I_D = 0.25 mA dc	$V_{GS(th)1}$	2.0	4.0	V dc
Gate current	3411	V_{GS} = ±20 V dc, V_{DS} = 0 Bias condition C	I_{GSS1}		±100	nA dc
Drain current	3413	V_{GS} = 0 V dc V_{DS} = 80 percent of rated V_{DS}; Bias condition C	I_{DSS1}		25	µA dc
Static drain to source on-state resistance	3421	V_{GS} = 10 V dc, pulsed (see 4.5.1) Condition A, I_D = rated I_{D2} (see 1.3)	$r_{DS(on)1}$			ohms
2N6756					0.18	
2N6758					0.4	
2N6760					1.0	
2N6762					1.5	
Static drain to source on-state resistance	3421	V_{GS} = 10 V dc, pulsed (see 4.5.1) Condition A, I_D = rated I_{D1} (see 1.3)	$r_{DS(on)2}$			ohms
2N6756					.21	
2N6758					.49	
2N6760					1.22	
2N6762					1.80	
Forward voltage (source drain diode)	4011	Pulsed (see 4.5.1) V_{GS} = 0 V, I_D = I_{D1}	V_{SD}			V
2N6756					1.8	
2N6758					1.6	
2N6760					1.5	
2N6762					1.4	

See footnote at end of table.

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TABLE I Group A inspection - Continued.

Inspection 1/	MIL STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 3						
High temperature operation:		$T_C = T_J = +125°C$				
Gate current	3411	Bias condition C $V_{GS} = \pm 20$ V dc $V_{DS} = 0$ V dc	I_{GSS2}		±200	nA dc
Drain current	3413	Bias condition C $V_{GS} = 0$ V dc V_{DS} = 100 percent of rated V_{DS}	I_{DSS2}		1.0	mA dc
		V_{DS} = 80 percent of rated V_{DS}	I_{DSS3}		0.25	mA dc
Static drain to source on-state resistance	3421	V_{GS} = 10 V dc pulsed (see 4.5.1) I_D = rated I_{D2}	$r_{DS(on)3}$			ohms
2N6756					0.34	
2N6758					0.8	
2N6760					2.2	
2N6762					3.3	
Gate to source voltage (threshold)	3404	$V_{DS} \geq V_{GS}$ I_D = 0.25 mA dc	$V_{GS(th)2}$	1.0		V dc
Low temperature operation:		$T_C = T_J = -55°C$				
Gate to source voltage (threshold)	3403	$V_{DS} \geq V_{GS}$ I_D = 0.25 mA dc	$V_{GS(th)3}$		5.0	V dc
Subgroup 4						
Switching time test	3472	I_D = rated I_{D1} V_{GS} = 10 V dc Gate drive impedance = 7.5Ω V_{DD} = 50 percent of $V_{BR(DSS)}$				
Turn-on delay time			$t_{d(on)}$			ns
2N6756					35	
2N6758					35	
2N6760					30	
2N6762					30	

See footnote at end of table.

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TABLE I. Group A inspection - Continued.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 4 - Continued						
Rise time			t_r			ns
2N6756					80	
2N6758					80	
2N6760					40	
2N6762					40	
Turn-off delay time			$t_{d(off)}$			ns
2N6756					60	
2N6758					60	
2N6760					80	
2N6762					80	
Fall time			t_f			ns
2N6756					45	
2N6758					40	
2N6760					35	
2N6762					30	
Subgroup 5						
Safe operating area test	3474	See figure 5 V_{DS} = 80 percent of rated $V_{BR(DSS)}$ V_{DS} = 200 V max t_p = 10 ms				
Electrical measurements		See table I, group A, subgroup 2				
Single pulse unclamped inductive switching 3/	3470	See 4.5.4 c = 0, 116 devices				
Electrical measurements		See table I, group A, subgroup 2				
Subgroup 6						
Not applicable						

See footnote at end of table.

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TABLE I. Group A inspection - Continued.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 7						
Gate charge	3471	Condition B				
On-state gate charge			$Q_{g(on)}$			nC
2N6756					35	
2N6758					39	
2N6760					39	
2N6762					40	
Gate to source charge			Q_{gs}			nC
2N6756					10	
2N6758					5.7	
2N6760					6.0	
2N6762					6.0	
Gate to drain charge			Q_{gd}			nC
2N6756					15	
2N6758					20	
2N6760					20	
2N6762					20	
Reverse recovery time	3473	di/dt = 100 A/μs $V_{DD} \leq 30$ V	t_{rr}			ns
2N6756		$I_D = I_{D1}$			300	
2N6758					500	
2N6760					700	
2N6762					900	

1/ For sampling plan, see MIL-S-19500.

2/ This test is required for the following end-point measurements only (not intended for screen 13):

JANS - group B, subgroups 3 and 4

JANTX and JANTXV - group B, subgroups 2 and 3;
group C, subgroup 6;
group E, subgroup 1

3/ This test is optional if performed as a 100% screen.

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TABLE II. Group E inspection (all quality levels) for qualification only

Inspection 1/	MIL-STD-750 Method	MIL-STD-750 Conditions	Qualification and large lot quality conformance inspection 1/
Subgroup 1			45 devices c = 1
Temperature cycle	1051	Condition G, 500 cycles	
Hermetic seal			
a. Fine leak b. Gross leak			
Electrical measurements		See table I, group A, subgroup 2	
Subgroup 2 2/			45 devices c = 1
Steady-state reverse bias	1042	Condition A, 1000 hours	
Electrical measurements		See table I, group A, subgroup 2	
Steady-state gate bias	1042	Condition B, 1000 hours	
Electrical measurements		See table I, group A, subgroup 2	
Subgroup 3			
Not applicable			
Subgroup 4			5 devices c = 0
Thermal resistance	3161	$R_{\theta JC}$ = 1.67°C/W max. See 4.5.2	
Subgroup 5			5 devices c = 0
Barometric pressure (reduced) 400 and 500 V only	1001	Test condition C $V_{ISO} = V_{DS}$ $I_{(ISO)}$ = .25 mA (max)	

1/ JANHC and JANKC devices are qualified in accordance with appendix H of MIL-S-19500.
2/ A separate sample for each test may be pulled.

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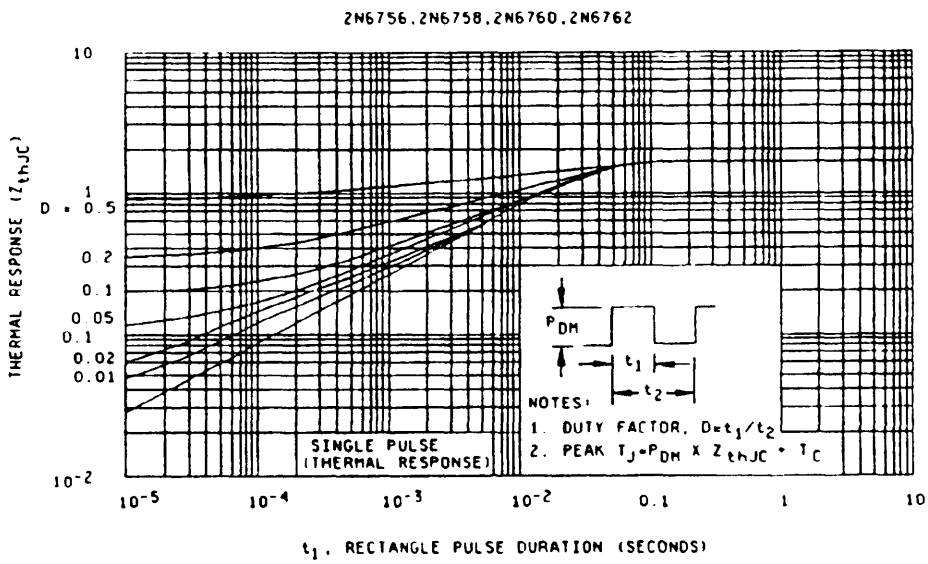


FIGURE 4. Thermal response curves.

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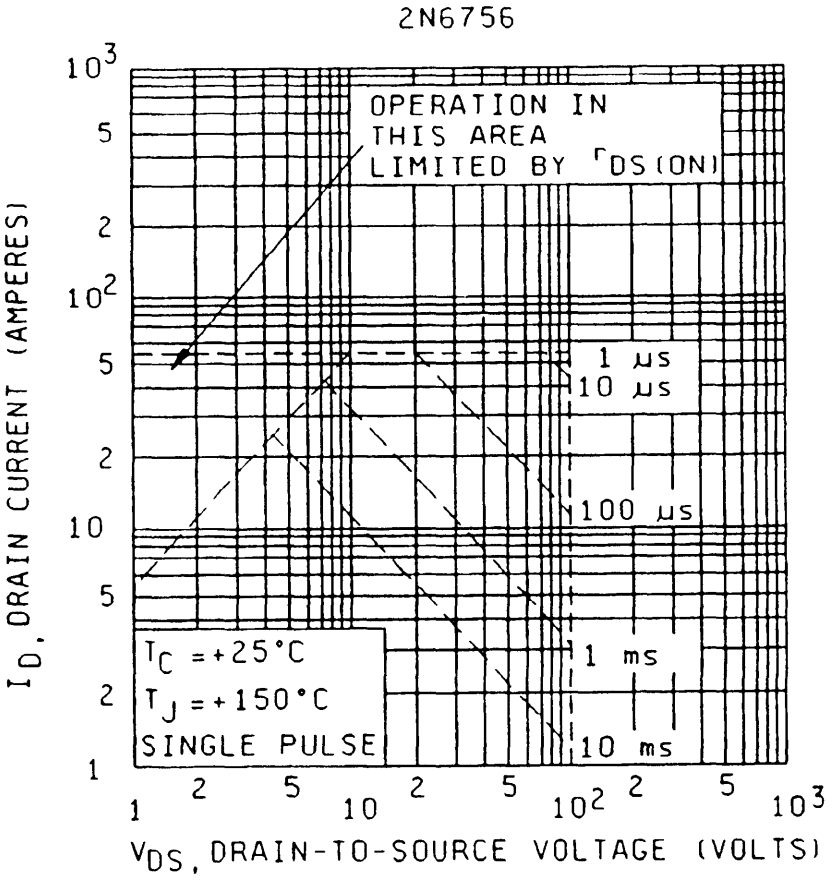


FIGURE 5. Safe operating area.

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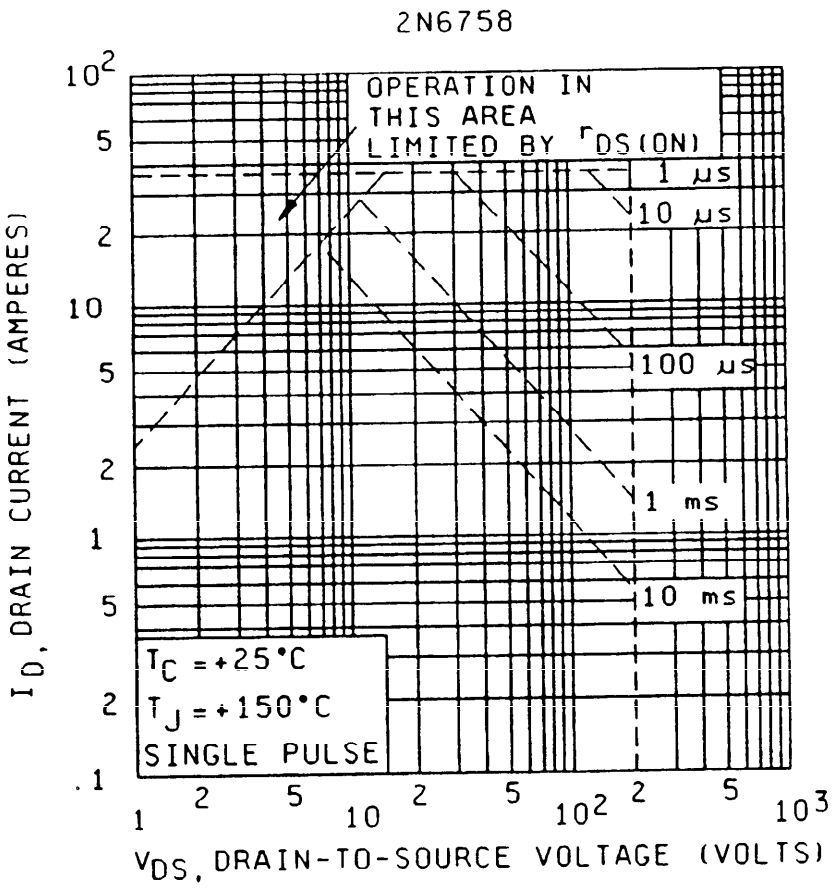


FIGURE 5. Safe operating area - Continued.

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2N6760

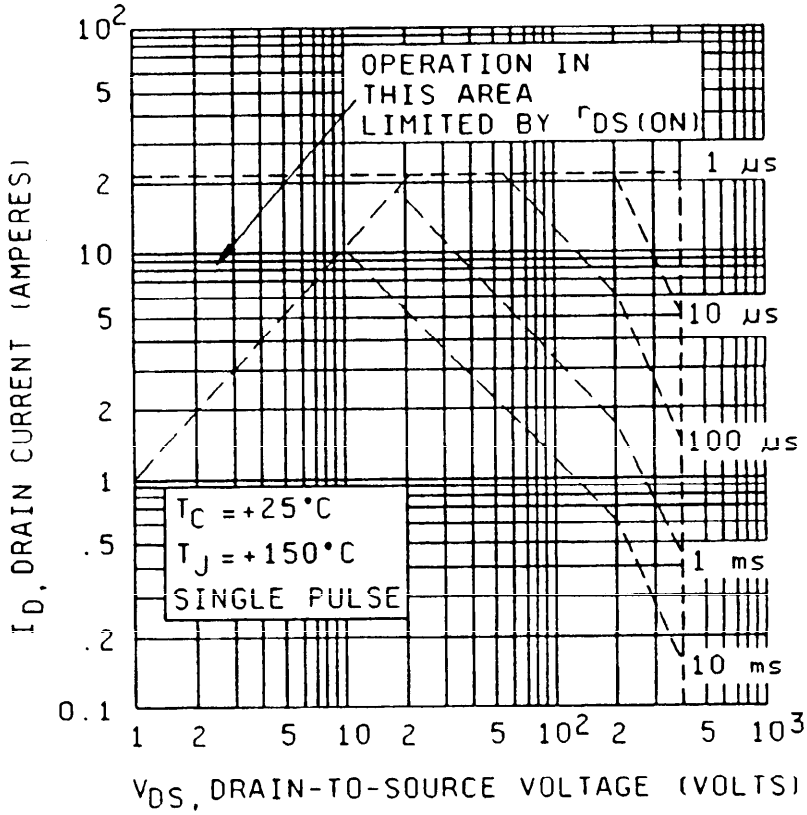


FIGURE 5. Safe operating area - Continued.

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2N6762

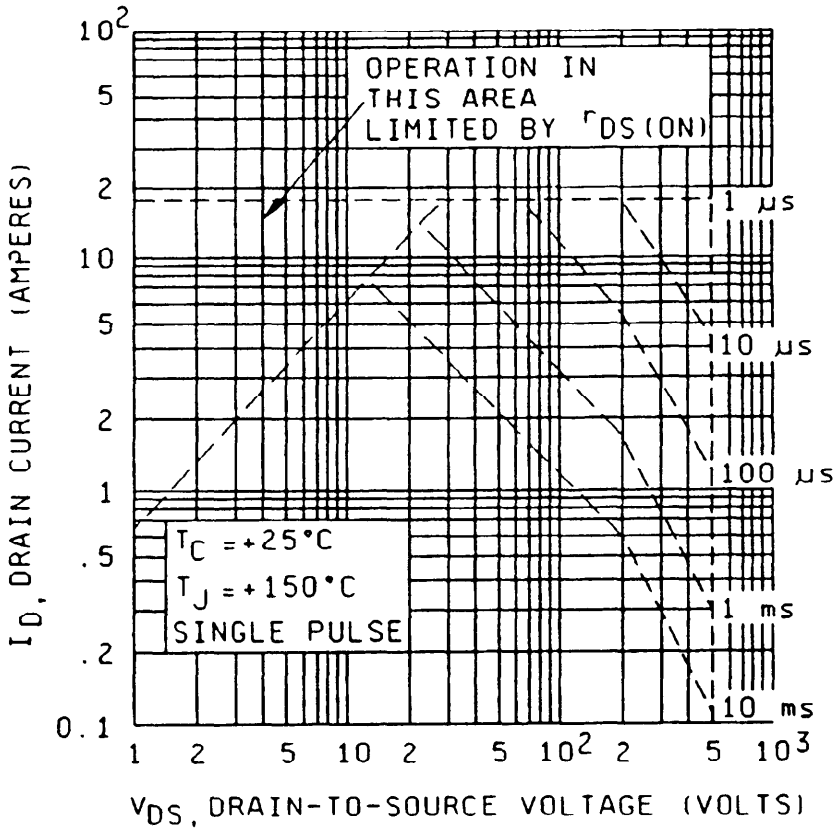


FIGURE 5. Safe operating area - Continued.

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5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-S-19500.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory).

6.1 Notes. The notes specified in MIL-S-19500 are applicable to this specification.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of the specification.

b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1).

c. Lead material and finish as specified (see 3.3.1).

d. For die acquisition, specify the JANHC or JANKC letter version (see figures 3, 4, and 5).

e. Type designation and product assurance level.

6.3 Substitution information. Devices covered by this specification are substitutable for the manufacturer's and user's Part or Identifying Number (PIN's). This information in no way implies that manufacturer's PIN's are suitable as a substitute for the military PIN's.

PIN	Manufacturer's CAGE code	Manufacturer's and user's PIN
2N6756	17856, 59993, 18722, 04713	IRF130
2N6758	17856, 59993, 18722, 04713	IRF230
2N6760	17856, 59993, 18722, 04713	IRF330
2N6762	17856, 59993, 18722, 04713	IRF430

6.4 Replacement data. JANTX devices shall be a direct replacement for JAN devices (example: JANTX2N6756 for JAN2N6756). Manufactures can no longer manufacture device types from this specification with JAN assurance level.

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6.5 Suppliers of JANHC and JANKC die. The qualified die suppliers with the applicable letter version (example JANHCA2N6756) will be identified on the QPL.

JANC ordering information		
PIN	Manufactures	
	59993	18722
2N6756	JANHCA6756 JANKCA6756	JANHCB6756 JANKCB6756
2N6758	JANHCA6758 JANKCA6758	JANKCB6758 JANKCB6758
2N6760	JANHCA6760 JANKCA6760	JANKCB6760 JANKCB6760
2N6762	JANHCA6762 JANKCA6762	JANHCB6762 JANKCB6762

6.6 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

CONCLUDING MATERIAL

Custodians:
Army - ER
Navy - EC
Air Force - 17
NASA - NA

Review activities:
Navy - TD
Air Force - 19, 70, 80

Preparing activity:
DLA - ES

(Project 5961-1681)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.

2. The submitter of this form must complete blocks 4, 5, 6, and 7.

3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-S-19500/542E	2. DOCUMENT DATE (YYMMDD) 94/08/17

3. DOCUMENT TITLE
SEMICONDUCTOR DEVICE, FIELD EFFECT TRANSISTORS, N-CHANNEL, SILICON, TYPES 2N6756, 2N6758, 2N6760, 2N6762 JANTX, JANTXV, JANS, JANHC, AND JANKC

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME Alan Barone	b. TELEPHONE (Include Area Code) (1) Commercial: 513-296-6048 (2) AUTOVON: 986-6048
c. ADDRESS (Include Zip Code) Defense Electronics Supply Center Attn: DESC-EIDT Dayton, Ohio 45444-5765	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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